

Chapter 4

Consumer and Firm Behavior: The Work-Leisure Decision and Profit Maximization

■ Teaching Goals

The microeconomic approach to macroeconomics stresses the notion that economy-wide events are the result of decisions made by individuals. People work so that they may afford to buy market goods. On the other hand, people generally prefer to work less rather than working more. Although discussions in the popular press often refer to the idea that spending is what drives the economy, an economy cannot produce unless people are willing to work. Therefore, the most basic macroeconomic decision is the decision to choose whether, and how much, to work. Production and willingness to work are intrinsically interconnected.

Students often believe that how much a person works is largely determined by the necessities of their circumstances. Students will report that they have to work to survive and pay tuition. Some might point out that some students need not work much or at all because their parents provide more support. However, circumstances need not dictate exactly how much they may choose to work. They may work less if they go to a less costly school. They may sometimes decide to switch to part-time student status and full-time work status if they find a high-paying job. A key message of this chapter is that choice is important and that choice is influenced by changes in circumstances.

This chapter demands the mastery of a large body of structure and yet provides little in the way of immediate insights. Students may need frequent assurances that the mastery of this material eventually pays big dividends in providing hope of understanding the phenomenon of business cycles. This is particularly important as this chapter lays critical foundations for the rest of the book: the use of microfoundations in macroeconomics. Students need to be able to justify macroeconomic relationships with microeconomic arguments, like in this chapter. This requires to some extent some boring drills that they will come to appreciate only later. If for many textbooks the strategy is to teach one chapter a week, spend more time on this one, especially if students have not yet mastered intermediate microeconomics.

Two key points of this chapter are the concepts of income and substitution effects. Often, students are perplexed at the amount of time spent on this material because nothing in practice is purely an income effect or a substitution effect. However, the two most basic insights of microeconomic analysis are that when we become more well-off we generally want more of everything and that we respond to price incentives at the margin.

■ Classroom Discussion Topics

Ask the students about their work choices and the choices of their parents, friends, and relatives. Does everyone work? Does everyone work the same amount of hours? Then ask the students for examples of the kinds of factors that lead people to work more or less. Try to elicit very specific examples. Then ask the students to categorize these factors that lead to more or less work. Some of these factors are actually the by-products of more complex decision making. For example, if they say that they work more or less because they go to school, point out that going to school is a choice. They may also point to circumstances like whether a married couple has children, and if so, the number of children and their ages. Point out that these events are also the results of other choices. Then ask the students to try to categorize the remaining factors as being either primarily an income effect or a substitution effect. Also compare labor choices across countries, as the Macroeconomics in Action feature does.

Ask the students to provide examples of factors other than more labor or capital that can allow some countries to be a lot more productive than others. What factors other than growth in capital and labor allow a given economy to produce more (or less) over time? Explain that these are the kinds of factors that we summarize by the concept of total factor productivity. Insist also on the concept of physical capital and what it measures and what it does not measure.

■ Outline

I. The Representative Consumer

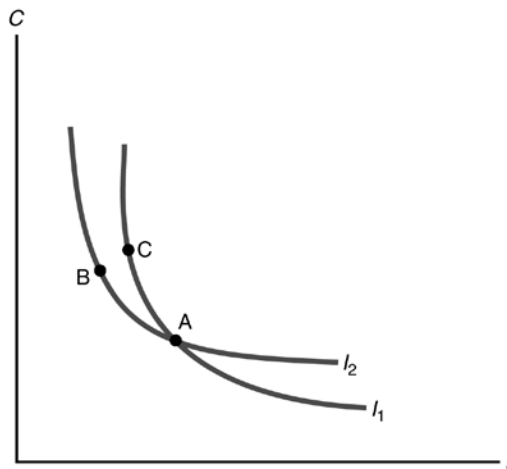
- A. Preferences
 - 1. Goods: The Consumption Good and Leisure
 - 2. The Utility Function
 - a. More Preferred to Less
 - b. Preference for Diversity
 - c. Normal Goods
 - 3. Indifference Curves
 - a. Downward Sloping
 - b. Convex to the Origin
 - 4. Marginal Rate of Substitution
- B. The Representative Consumer's Budget Constraint
 - 1. Competitive Behavior, Price-takers, Barter Economy, The Time Constraint
 - 2. Real Disposable Income
 - 3. The Budget Constraint Equations
 - 4. A Graphical Representation
- C. Optimization
 - 1. Rational Behavior
 - 2. The Optimal Consumption Bundle
 - 3. Marginal Rate of Substitution $\square$ Relative Price
 - 4. A Graphical Representation
- D. Comparative Statics Experiments
 - 1. Changes in Dividends and Taxes: Pure Income Effect
 - 2. Changes in the Real Wage: Income and Substitution Effects

II. The Representative Firm

- A. The Production Function
 - 1. Constant Returns to Scale
 - 2. Monotonicity
 - 3. Declining MP_N
 - 4. Declining MP_K
 - 5. Changes in Capital and MP_N
- B. Profit Maximization
 - 1. Profits = Total Revenue – Total Variable Costs
 - 2. Marginal Product of Labor = Real Wage
 - 3. Labor Demand

■ Solutions to End-of-Chapter Problems

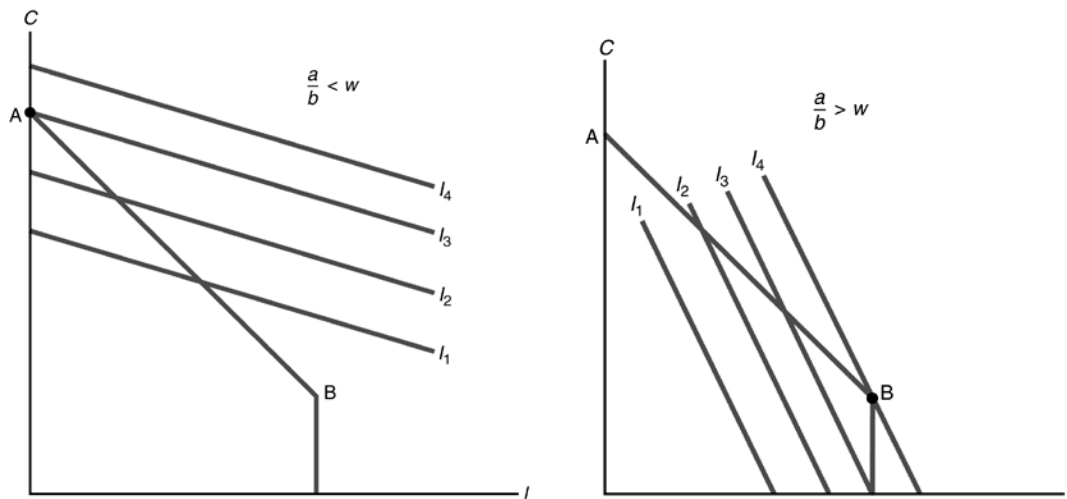
1. Consider the two hypothetical indifference curves in the figure below. Point A is on both indifference curves, I_1 and I_2 . By construction, the consumer is indifferent between A and B, as both points are on I_2 . In like fashion, the consumer is indifferent between A and C, as both points are on I_1 . But at point C, the consumer has more consumption and more leisure than at point B. As long as the consumer prefers more to less, he or she must strictly prefer C to A. We therefore contradict the hypothesis that two indifference curves can cross.



2. $u = al + bC$
 - (a) To specify an indifference curve, we hold utility constant at $\bar{u}$. Next rearrange in the form:

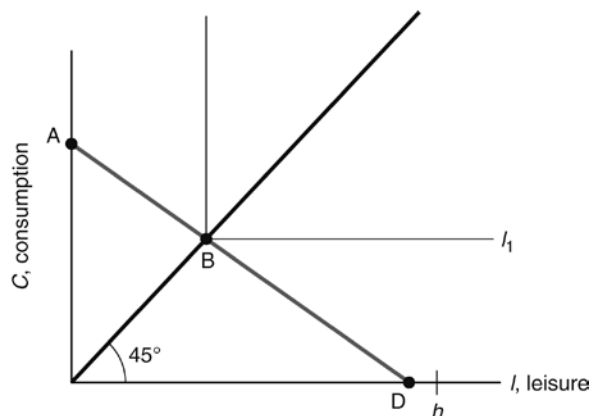
$$C = \frac{\bar{u}}{b} - \frac{a}{b}l$$

Indifference curves are therefore linear with slope, $-a/b$, which represents the marginal rate of substitution. There are two main cases, according to whether $a/b > w$ or $a/b < w$. The top panel of the left figure below shows the case of $a/b < w$. In this case the indifference curves are flatter than the budget line and the consumer picks point A, at which $l = 0$ and $C = wh + \pi - T$. The right figure shows the case of $a/b > w$. In this case the indifference curves are steeper than the budget line, and the consumer picks point B, at which $l = h$ and $C = \pi - T$. In the coincidental case in which $a/b = w$, the highest attainable indifference curve coincides with the budget line, and the consumer is indifferent among all possible amounts of leisure and hours worked.



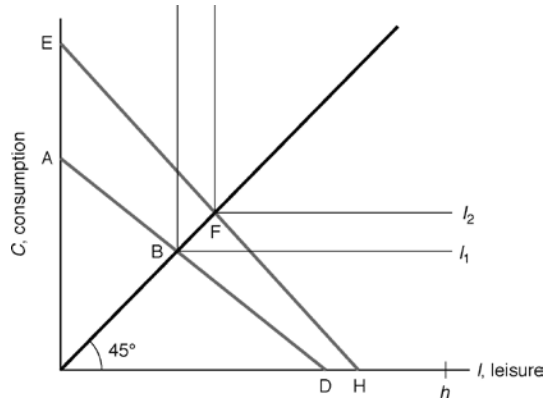
- (b) The utility function in this problem does not obey the property that the consumer prefers diversity, and is therefore not a likely possibility.
- (c) This utility function does have the property that more is preferred to less. However, the marginal rate of substitution is constant, and therefore this utility function does not satisfy the property of diminishing marginal rate of substitution.
3. (a) Using the formulas in the example from the textbook, one obtains:

$$l = C = (0.75 \times 16 - 0.8 - 6)/(1 + 0.75) = 3.89$$



Given the numbers given, we can precisely determine the coordinates of the points in the figure above: A is (0,6.8), B is (3.89,3.89), D is (9,0,0), with the slope of ABD being - 0.75.

- (b) With the new wage, we obtain: $l = C = (1.5 \times 16 - 0.8 - 6)/(1 + 1.5) = 7.52$



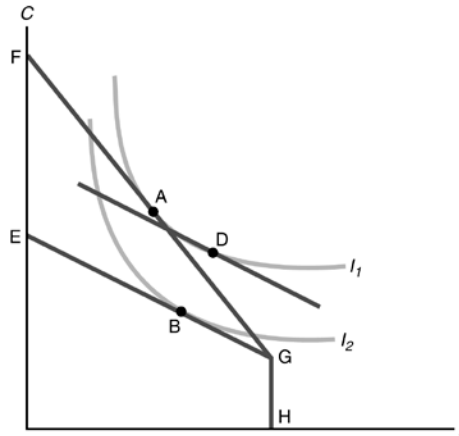
where A, B and D have the same coordinates as above, and E is (0, 12.8), F is (7.52, 7.52), H is (12.53,0), and the slope of EFH is - 1.5. As there are no substitution effects when goods are perfect complements, the entire move from point B to point F is due to the income effect.

4. When the government imposes a proportional tax on wage income, the consumer's budget constraint is now given by:

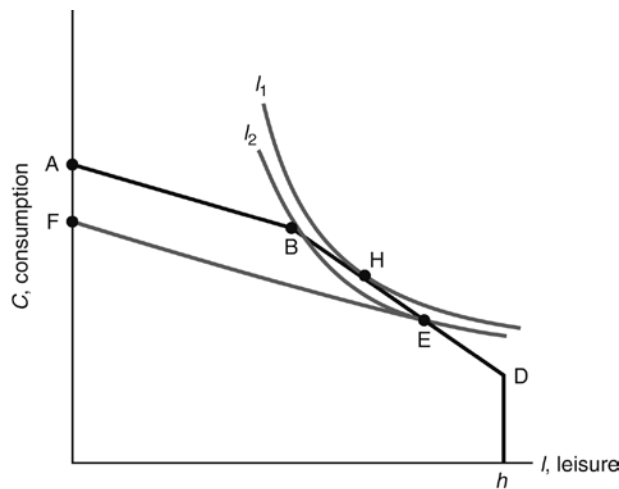
$$C = w(1-t)(h-l) + \pi - T,$$

where t is the tax rate on wage income. In the figure below, the budget constraint for $t \leq 0$, is FGH. When $t > 0$, the budget constraint is EGH. The slope of the original budget line is $-w$, while the slope of the new budget line is $-(1-t)w$. Initially the consumer picks the point A on the original budget line. After the tax has been imposed, the consumer picks point B. The substitution effect of the imposition of the tax is to move the consumer from point A to point D on the original indifference curve. The point D is at the tangency point of indifference curve, I_1 , with a line segment that is parallel to EG. The pure substitution effect induces the consumer to reduce consumption and increase leisure (work less).

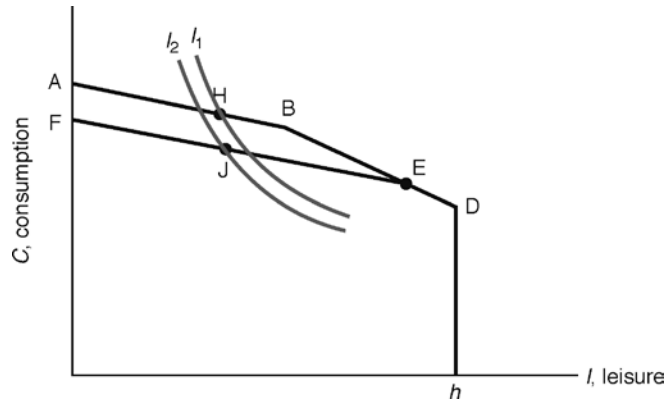
The tax also makes the consumer worse off, in that he or she can no longer be on indifference curve, I_1 , but must move to the less preferred indifference curve, I_2 . This pure income effect moves the consumer to point B, which has less consumption and less leisure than point D, because both consumption and leisure are normal goods. The net effect of the tax is to reduce consumption, but the direction of the net effect on leisure is ambiguous. The figure shows the case in which the substitution effect on leisure dominates the income effect. In this case, leisure increases and hours worked fall. Although consumption must fall, hours worked may rise, fall, or remain the same.



5. The budget constraint has a kink due to the tax deduction and is represented in the following figures by ABDh. Reducing the tax deduction pushes the budget constraint to FEDh.

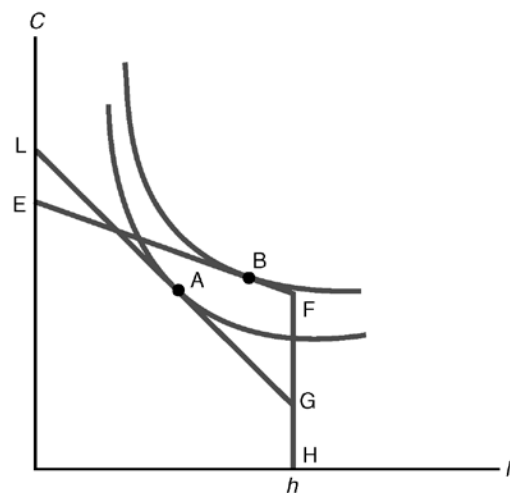
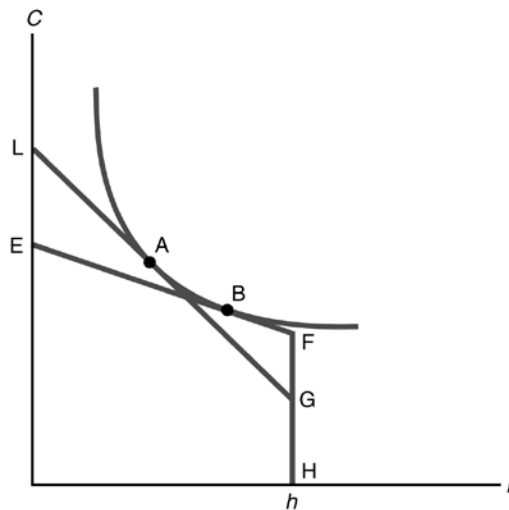


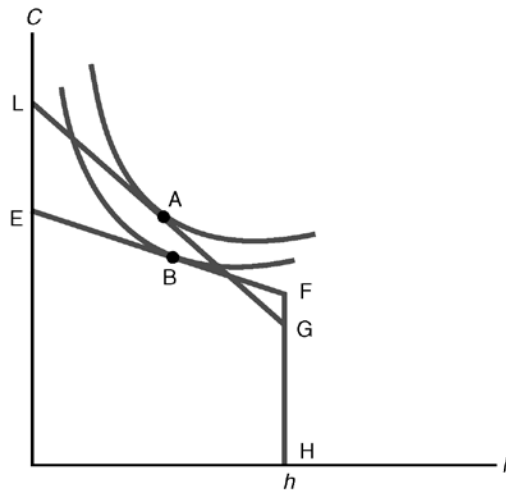
First consider a consumer who does not pay taxes. In the old regime, he would have an optimal bundle somewhere between B and D. Two things can happen. If the bundle is between E and D, there is no change. If it is between B and E, say at H, then the household will reoptimize with the new tax deduction. The new bundle is then either somewhere between E and F, and the MRS equals $w(1 - t)$. Or we obtain a corner solution at E, and the MRS is somewhere between w and $w(1 - t)$. The move from H to E is due to the income effect, and if there is an optimal strictly between E and F, the move from E to that point is due to the substitution effect.



For a consumer who pays taxes, his wage, and thus his MRS does not change. Thus the move from H to J is a pure negative income effect.

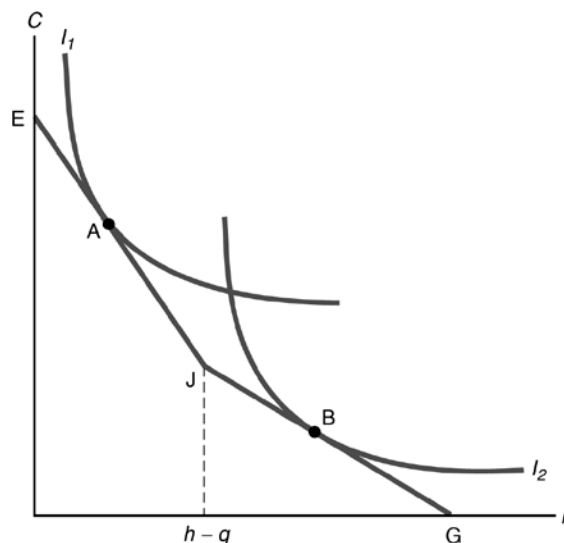
6. The increase in dividend income shifts the budget line upward. The reduction in the wage rate flattens the budget line. One possibility is depicted in the figures below. The original budget constraint HGL shifts to HFE. There are two income effects in this case. The increase in dividend income is a positive income effect. The reduction in the wage rate is a negative income effect. The drawing in the top figure shows the case where these two income effects exactly cancel out. In this case we are left with a pure substitution effect that moves the consumer from point A to point B. Therefore, consumption falls and leisure increases. As leisure increases, hours of work must fall. The middle figure shows a case in which the increase in dividend income, the distance GF, is larger and so the income effect is positive. The consumer winds up on a higher indifference curve, leisure unambiguously increases, and consumption may either increase or decrease. The bottom figure shows a case in which the increase in dividend income, the distance GF, is smaller and so the income effect is negative. The consumer winds up on a lower indifference curve, consumption unambiguously decreases, and leisure may either increase or decrease.





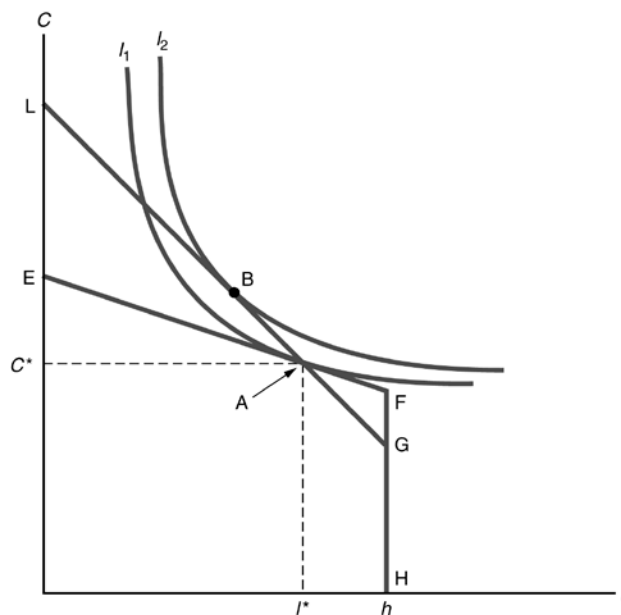
7. This problem introduces a higher, overtime wage for hours worked above a threshold, q . This problem also abstracts from any dividend income and taxes.

(a) The budget constraint is now EJG in the figure below. The budget constraint is steeper for levels of leisure less than $h - q$, because of the higher overtime wage. The figure depicts possible choices for two different consumers. Consumer #1 picks point A on her indifference curve, I_1 . Consumer #2 picks point B on his indifference curve, I_2 . Consumer #1 chooses to work overtime; consumer #2 does not.



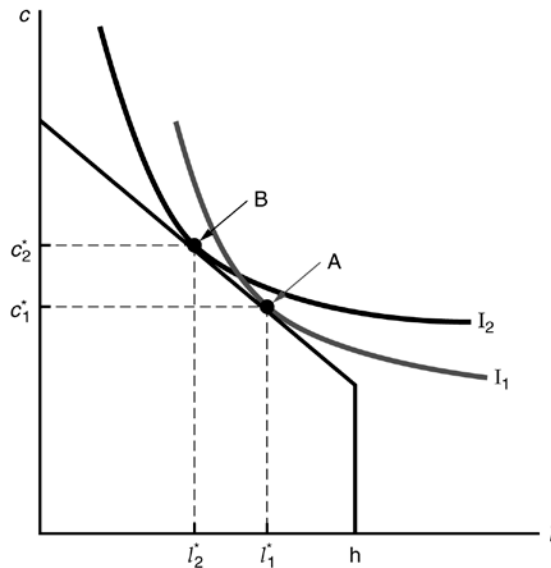
- (b) The geometry of the figure above makes it clear that it would be very difficult to have an indifference curve tangent to EJG close to point J . In order for this to happen, an indifference curve would need to be close to right angled as in the case of pure complement. It is unlikely that consumers wish to consume goods and leisure in fixed proportions, and so points like A and B are more typical. For any other allowable shape for the indifference curve, it is impossible for point J to be chosen.

- (c) An increase in the overtime wage steepens segment EJ of the budget constraint, but has no effect on the segment JG. For an individual like consumer #2, the increase in the overtime wage has no effect up until the point at which the increase is large enough to shift the individual to a point like point A. Consumer #2 receives no income effect because the income effect arises out of a higher wage rate on inframarginal units of work. An individual like consumer #1 has the traditional income and substitution effects of a wage increase. Consumer #1 increases her consumption, but may either increase or reduce hours of work according to whether the income effect outweighs the substitution effect.
8. Lump-sum Tax vs. Proportional Tax. Suppose that we start with a proportional tax. Under the proportional tax the consumer's budget line is EFH in the figure below. The consumer chooses consumption, C^* , and leisure, l^* , at point A on indifference curve I_1 . A shift to a lump-sum tax steepens the budget line. The absolute value of the slope of the budget line is $(1-t)w$, and t has fallen to zero. The imposition of the lump-sum tax shifts the budget line downward in a parallel fashion. By construction, the lump-sum tax must raise the same amount of revenue as the proportional tax. The consumer must therefore be able to continue to consume C^* of the consumption good and l^* of leisure after the change in tax collection. Therefore, the new budget line must also pass through point A. The new budget line is labeled LGH in the figure below. With the lump-sum tax, the consumer can do better by choosing point B, on the higher indifference curve, I_2 . Therefore, the consumer is clearly better off. We are also assured that consumption will be greater at point B than at point A, and that leisure will be smaller at point B than at point A.



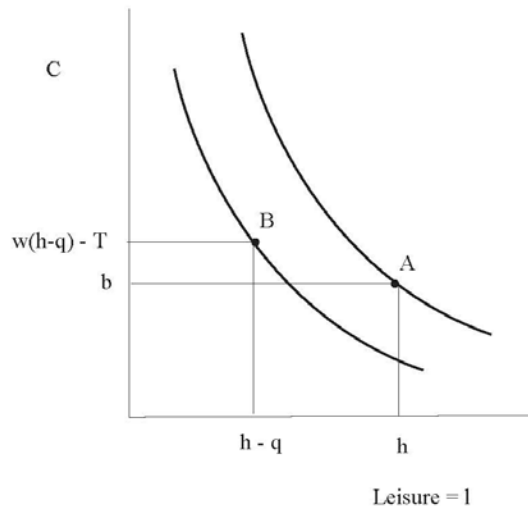
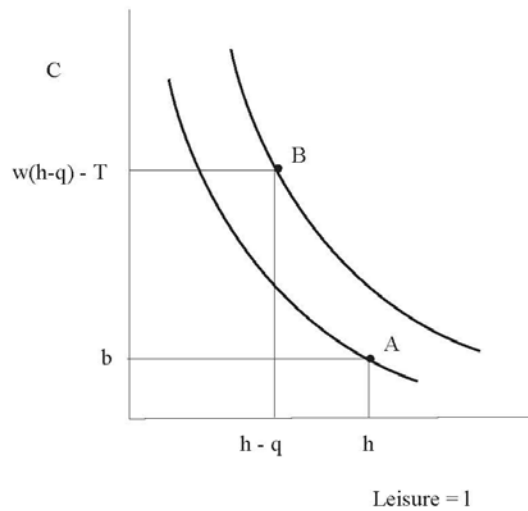
9. Leisure represents all time used for nonmarket activities. If the government is now providing for some of those, like providing free child care, households will take advantage of such a program, thereby allowing more time for other activities, including market work. Concretely, this translates in a change of preferences for households. For the same amount of consumption, they are now willing to work more, or in other words, they are willing to forego some additional leisure. On the

figure below, the new indifference curve is labeled I_2 . It can cross indifference curve I_1 because preferences, as we measure them here, have changed. The equilibrium basket of goods for the household now shifts from A to B. This leads to reduced leisure (from l_1^* to l_2^*), and thus increased hours worked, and increased consumption (from C_1^* to C_2^*) thanks to higher labor income at the fixed wage.



10. Supposing that the only options open to the consumer are working q hours and paying a tax T , or working zero hours and receiving an unemployment insurance benefit b , consumption will be $w(h-q)-T$ if the consumer works, and b if the consumer decides not to work. Then, either the consumer prefers not to work, as in the Figure 10.1, where the highest indifference curve is achieved at point A rather than at point B, or the consumer prefers to work, as in Figure 10.2. There is also another case where the consumer is just indifferent between working and not working, but that case is not important.

- a) Think of the economy as consisting of many consumers, some of whom are in a situation as in the Figure 10.1 and some as in Figure 10.2. Some consumers do not work, and some choose to work. If the wage goes up, then that will make working preferable for some consumers who formerly did not choose to work. An increase in the wage will not discourage anyone from working, but those who were working already will not choose to vary hours of work (they cannot). But total employment in the economy will increase, as now more people are working. With the constraint on hours of work, there are no issues related to income and substitution effects. A higher wage always increases the total quantity of labor supplied.

**Figure 10.1****Figure 10.2**

- b) Similar to part (a), if the unemployment insurance benefit increases, this will make not working preferable to some consumers who were formerly working, and employment will fall. An increase in the unemployment insurance benefit unequivocally reduces the quantity of labor supplied.
11. The firm chooses its labor input, N^d , so as to maximize profits. When there is no tax, profits for the firm are given by

$$\pi = zF(K, N^d) - wN^d.$$

That is, profits are the difference between revenue and costs. In the top figure on the following page, the revenue function is $zF(K, N^d)$ and the cost function is the straight line, wN^d . The firm maximizes profits by choosing the quantity of labor where the slope of the revenue function equals the slope of the cost function:

$$MP_N = w.$$

The firm's demand for labor curve is the marginal product of labor schedule in the bottom figure on the following page.

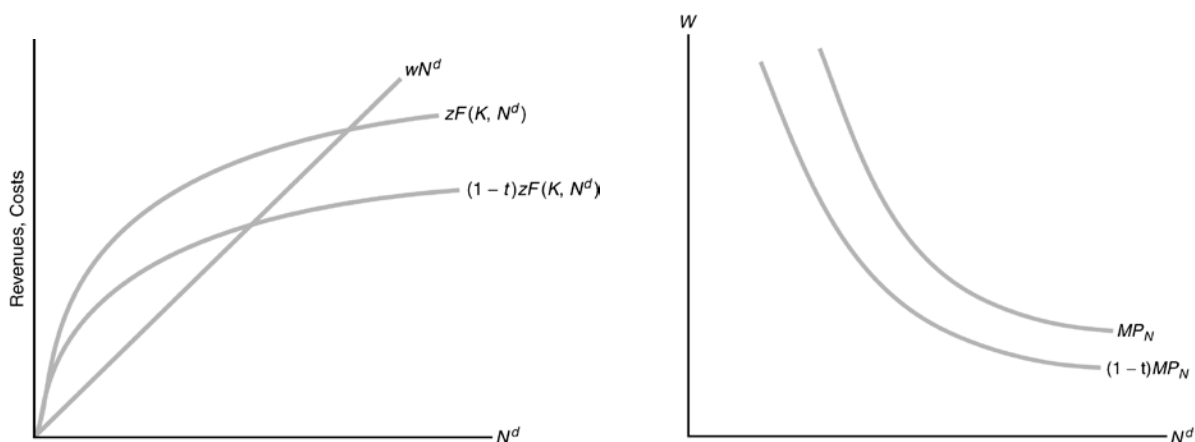
With a tax that is proportional to the firm's output, the firm's profits are given by:

$$\begin{aligned}\pi &= zF(K, N^d) - wN^d - tzF(K, N^d) \\ &= (1-t)zF(K, N^d),\end{aligned}$$

where the term $(1-t)zF(K, N^d)$ is the after-tax revenue function, and as before, wN^d is the cost function. In the top figure below, the tax acts to shift down the revenue function for the firm and reduces the slope of the revenue function. As before, the firm will maximize profits by choosing the quantity of labor input where the slope of the revenue function is equal to the slope of the cost function, but the slope of the revenue function is $(1-t)MP_N$, so the firm chooses the quantity of labor where

$$(1-t)MP_N = w.$$

In the bottom figure below, the labor demand curve is now $(1-t)MP_N$, and the labor demand curve has shifted down. The tax acts to reduce the after-tax marginal product of labor, and the firm will hire less labor at any given real wage.



12. The firm chooses its labor input N^d so as to maximize profits. When there is no subsidy, profits for the firm are given by

$$\pi = zF(K, N^d) - wN^d.$$

That is, profits are the difference between revenue and costs. In the top figure on the following page the revenue function is $zF(K, N^d)$ and the cost function is the straight line, wN^d . The firm maximizes profits by choosing the quantity of labor where the slope of the revenue function equals the slope of the cost function:

$$MP_N = w.$$

The firm's demand for labor curve is the marginal product of labor schedule in the bottom figure below.

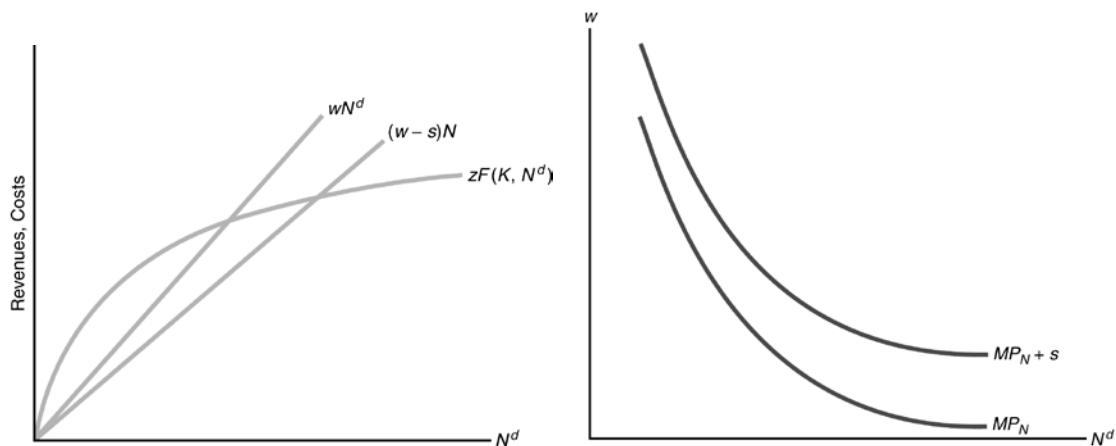
With an employment subsidy, the firm's profits are given by:

$$\pi = zF(K, N^d) - (w - s)N^d$$

where the term $zF(K, N^d)$ is the unchanged revenue function, and $(w - s)N^d$ is the cost function. The subsidy acts to reduce the cost of each unit of labor by the amount of the subsidy, s . In the top figure below, the subsidy acts to shift down the cost function for the firm by reducing its slope. As before, the firm will maximize profits by choosing the quantity of labor input where the slope of the revenue function is equal to the slope of the cost function, $(w - s)$, so the firm chooses the quantity of labor where

$$MP_N = w - s.$$

In the bottom figure below, the labor demand curve is now $MP_N + s$, and the labor demand curve has shifted up. The subsidy acts to reduce the marginal cost of labor, and the firm will hire more labor at any given real wage.



13. In Figure 13.1, given the minimum quantity of employment that the firm requires to operate, the production function (identical to the total revenue function) follows ABD, and then continues

along the same production function we would have without the minimum quantity of employment. The firm maximizes profits, which implies that, if the market wage rate is greater than w^* the firm will earn negative profits for any quantity of labor input greater than or equal to N^* . Therefore, if $w > w^*$ then $N^d = 0$. If the real wage rate is less than or equal to w^* (so that the firm can earn positive profits for at least some positive quantities of labor input), but larger than MP_N^* (the marginal product of labor when $N^d = N^*$), then the firm will choose $N^d = N^*$ to maximize profits, as is the case in Figure 13.1 when $w = w_1$. That is, choosing $N^d = N^*$ is better than choosing $N^d = 0$ in this case, because the firm will earn positive profits rather than zero profits. However, if the firm increases the labor input above N^* , this will just reduce profits, as $w > MP_N$ when $N^d \geq N^*$. Now, if $w \leq MP_N^*$, then the minimum quantity of employment does not make any difference for what the firm chooses to do. The firm sets N^d so that $w = MP_N$, and the firm will choose $N^d = N^*$, as in Figure 4.14 when the firm faces a market wage w_2 and chooses $N^d = N_2$. Therefore, the labor demand curve is as depicted in Figure 13.2. The interesting feature of the firm's behavior is that labor demand does not change smoothly in response to the real wage. There is a critical wage rate w^* at which the firm is willing to start up, and at that wage it will actually hire more labor than would an identical firm that did not face a minimum employment constraint. In reality, firms may face constraints like this, for example a restaurant needs at least one cook, one waiter, and one cashier to operate, and this may cause employment to increase and decrease by larger amounts in response to changes in market wages than would otherwise be the case.

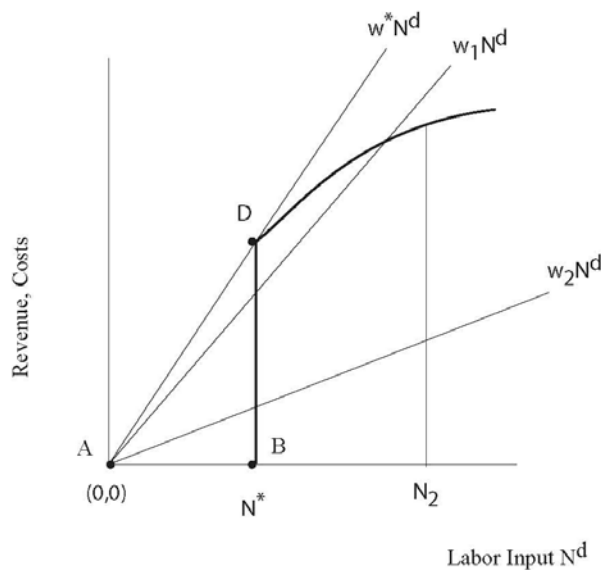


Figure 13.1

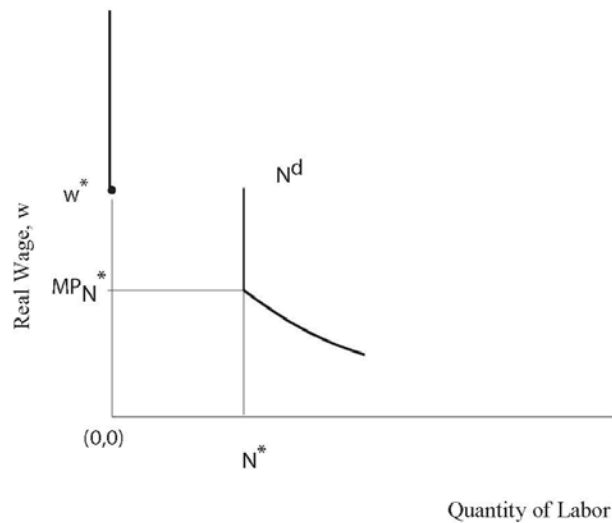
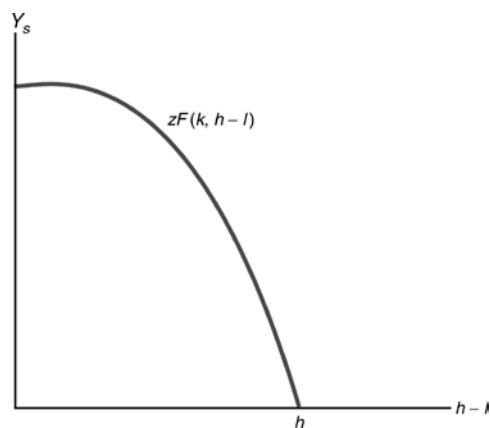


Figure 13.2

14. The level of output produced by one worker who works $h - l$ hours is given by

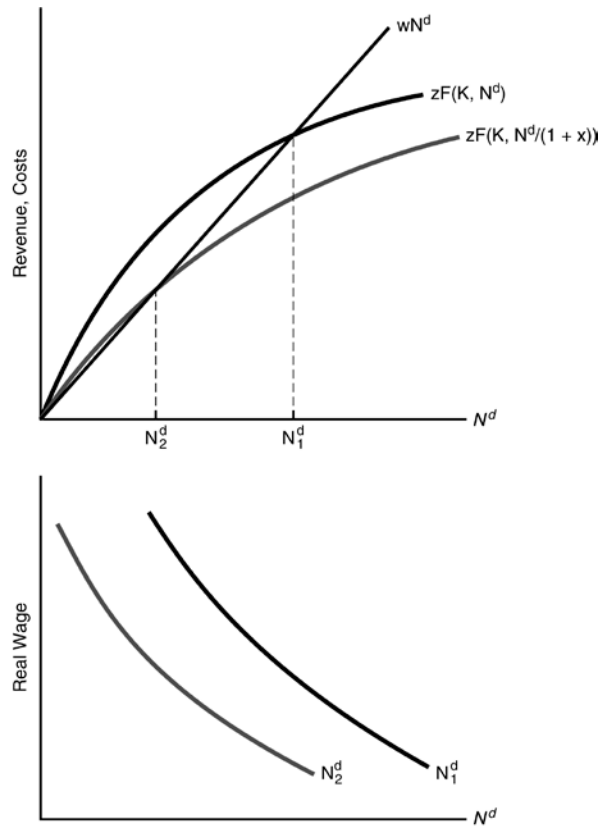
$$Y_s = zF(K, h - l).$$

This equation is plotted in the figure below. The slope of this production possibilities frontier is simply $-MP_N$.



15. As the firm has to internalize the pollution, it realizes that labor is less effective than it previously thought. It now needs to hire $N(1 - \alpha)$ workers where N were previously sufficient. This is qualitatively equivalent to a reduction of z , total factor productivity. The figure below highlights

the resulting outcome: the firm now hires fewer people for a given wage and thus its labor demand is reduced.



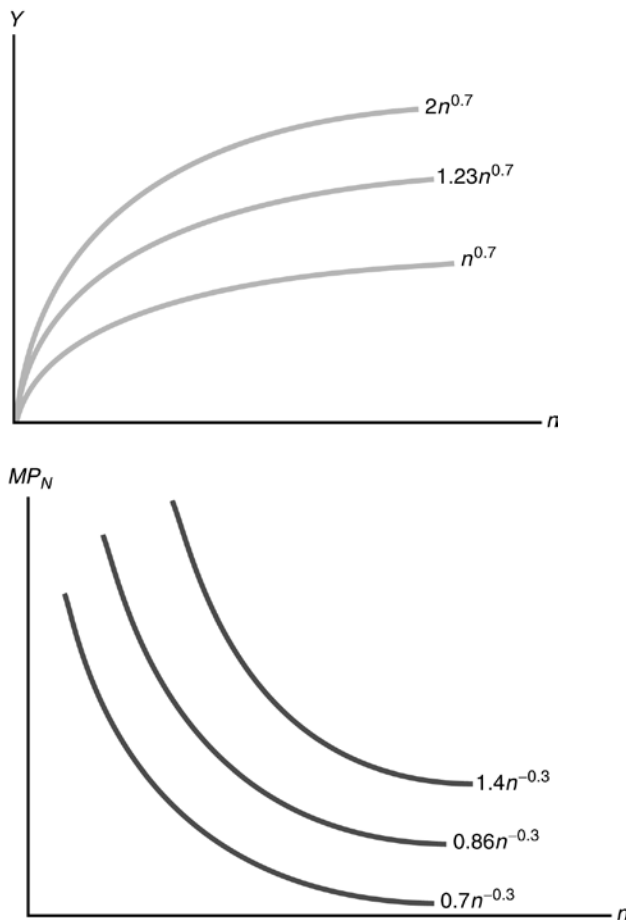
16. $Y = zK^{0.3}n^{0.7}$

- (a) $Y = n^{0.7}$. See the top figure below. The marginal product of labor is positive and diminishing.
- (b) $Y = 2n^{0.7}$. See the figures below.
- (c) $Y = 2^{0.3}n^{0.7} \approx 1.23n^{0.7}$. See the figures below.
- (d) See the bottom figure below.

$$z = 1, K = 1 \Rightarrow MP_N = 0.7n^{-0.3}$$

$$z = 2, K = 1 \Rightarrow MP_N = 1.4n^{-0.3}$$

$$z = 1, K = 2 \Rightarrow MP_N = 2^{0.3} \times 0.7n^{-0.3} \approx 0.86n^{-0.3}$$



17. a) The production function, for fixed K , is shown in Figure 17.1. For any wage w , profits for the firm will increase with N , and there is no limit on how much labor the firm wants to hire. The firm maximizes profits by hiring an infinite amount of labor.

- b) This presents problems for competitive equilibrium because supply could never equal demand in equilibrium, as the quantity of labor supplied could not be infinite. Essentially, competitive equilibrium is the wrong modeling approach if there are increasing returns to scale. A more appropriate approach would be to assume a monopoly producer, or at least an oligopoly where firms behave strategically instead of as price takers.

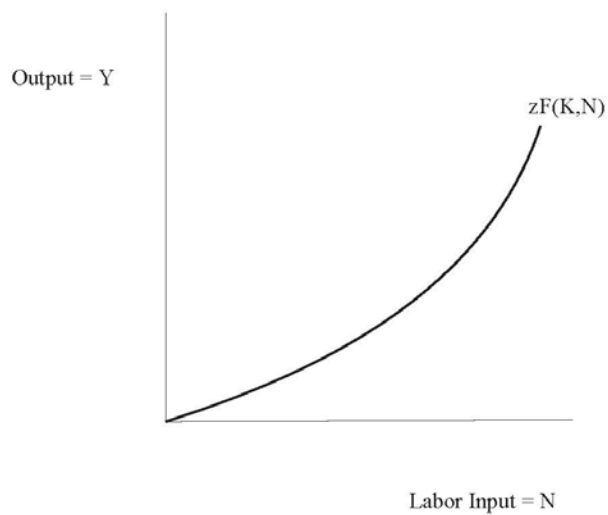


Figure 17.1